ECG-AI: An App for ECG Interpretation in Remote Regions of the KR

Relevance of the Problem

Cardiovascular diseases (CVD) are the leading cause of death in Kyrgyzstan: they consistently account for **50–52% of all deaths**, which amounts to **17–20 thousand people every year** (~50 deaths per day). In mortality from ischaemic heart disease, the Kyrgyz Republic (KR) ranks **3rd among women** and **5th among men** out of the 51 countries of the WHO European Region. More than **65% of the population** lives in rural areas, where the shortage of staff and technology is felt most acutely.

Key Data

Indicator	Value	Source
Share of CVD in the structure of mortality	50–52%	MSS KR, 2022–2026
CVD deaths per year	~19,000	IJMEDRI, 2024
Mortality per 100,000 population	234.5	European Heart Journal, 2024
Share of population with arterial hypertension	34–37%	NCCIM (National Center of Cardiology and Internal Medicine)
Share of rural population	65.1%	NSC (National Statistical Committee), 2023
Adults aged 40–64 with a 10-year CVD risk >30%	every 5th	World Bank PHC Report

The Problem: Staff Shortage and Lack of ECG Expertise at the Primary Care Level

- **Shortage of cardiologists.** Feldshers and general practitioners (GPs) record ECGs but lack sufficient training to recognise life-threatening arrhythmias and acute coronary syndromes (ACS). The mountain regions (Naryn, Talas, Batken, Alay) are critically under-served by specialists.
- **Diagnostic delay = death.** In acute myocardial infarction, every 30 minutes of delay increases mortality by 7–8%. Telemedicine transmission of ECGs is effectively unavailable in most FAPs (feldsher-midwife posts).
- **The equipment is there — the interpretation is not.** Under the primary health care (PHC) reforms, ECG machines were purchased for rural facilities without a decision-support system.
- **Digital medicine is under-used.** The WHO and the World Bank state explicitly: telemedicine remains an "untapped lever" in the remote communities of the KR.

The Technological Opportunity Already Exists

Deep learning models based on the MIT-BIH Arrhythmia Database achieve **98–99.4% accuracy** in arrhythmia classification. Algorithms for digitising paper ECGs from a smartphone photo process an image in **less than 7 seconds** while being robust to image defects (folds, stains, perspective distortion). This makes the task technically feasible as a research project.

European Heart Journal, 2024 · IJMEDRI, 2024 · Nature — ECG digitization, 2025 · Nature — Deep learning ECG, 2024

Project Concept

ECG-AI KG is a mobile application with an AI backend for the automatic interpretation of paper ECG strips from a phone photo. Target users: feldshers, nurses and GPs in the FAPs/CFMs (Centres of Family Medicine) of remote districts of the KR.

Project Goal

To reduce the time from recording an ECG to a clinical decision on patient management **from several hours to 1–2 minutes** in facilities without a cardiologist, thereby increasing survival in ACS, life-threatening arrhythmias and other acute conditions.

User Scenario

1. The feldsher records an ECG on the machine and takes a photo of the strip with a smartphone.
2. Uploads the photo to the ECG-AI KG app.
3. The AI model digitises the signal and classifies the pathology.
4. On screen — the interpretation: normal / suspected pathology + type + urgency level.
5. The app provides a brief algorithm of next steps (emergency evacuation / observation / consultation).

Key Principles

- **Offline mode** — works without internet (critical for mountain FAPs).
- **Support, not replacement** — the result is presented as "interpretation assistance", not a diagnosis; the final decision rests with the physician.
- **Simple interface** — designed for a user with an average level of digital literacy.
- **Bilingual** — Kyrgyz and Russian interface.
- **Logging** — all cases are saved for subsequent audit and model fine-tuning.

TS (Technical Specification): System Description

The system consists of three components: a mobile application (client), an AI processing server and an administrative panel.

System Components

Component	Description
Mobile application	Android/iOS, offline mode (lite model), ECG photo upload, result display
AI backend (cloud)	Image processing server: digitisation → classification → report generation
Administrative panel	Case monitoring, user management, data export for the MoH KR (Ministry of Health of the KR)

Input Data

- A photo of a paper ECG strip (12-lead or abbreviated format) taken with a standard smartphone camera.
- Metadata: patient age and sex, facility, date/time.

Output Data

- **Interpretation**: normal / suspected pathology (with the type indicated).
- **Urgency category**: 🔴 emergency / 🟡 observation / 🟢 scheduled consultation.
- **List of detected patterns**: e.g. "Sinus tachycardia", "Signs of ACS with ST elevation", "Atrial fibrillation".
- **Brief action algorithm**: 3–5 steps for medical staff.
- **Visualisation**: highlighting of abnormal segments on the digitised ECG.

Target Pathologies (MVP)

In the first phase, the system recognises priority life-threatening conditions:

1. Acute coronary syndrome (ST elevation / ST depression)
2. Atrial fibrillation and flutter
3. Ventricular tachycardias / ventricular fibrillation
4. Complete AV block (3rd degree)
5. Sinus tachy- and bradyarrhythmias
6. Bundle branch blocks (LBBB / RBBB)

TS: Architecture and Technology Stack

Technology Stack

Layer	Technology	Rationale
Mobile application	Flutter (Android + iOS)	Single codebase, offline support

Layer	Technology	Rationale
ECG image digitisation	YOLOv11 (segmentation) + OpenCV	SOTA for paper ECG digitisation, open-source
Arrhythmia classification	CNN + BiLSTM (1D signal)	98–99.4% accuracy on MIT-BIH; combines spatial and temporal patterns
Training dataset	MIT-BIH Arrhythmia Database (PhysioNet)	Gold standard, 48 recordings, annotated by two cardiologists
Additional dataset	PTB-XL (21,837 ECGs)	Extension to 12-lead and ACS
Backend API	Python (FastAPI)	Fast development, native DL stack
Cloud	Google Cloud / local MoH server	Flexibility; on-premise deployment option for data sovereignty
Offline lite model	TensorFlow Lite	Basic classification without internet
Administrative panel	React + PostgreSQL	Standard web stack

ECG Image Processing Pipeline

1. **Image preprocessing** — perspective correction, shadow removal, alignment to the paper grid (OpenCV).
2. **Curve segmentation** — extracting the ECG signal from the background (YOLOv11 instance segmentation).
3. **Digitisation** — converting pixels into a numerical time series calibrated against the standard grid (25 mm/s, 10 mm/mV).
4. **Classification** — CNN + BiLSTM on the 1D signal, trained on MIT-BIH + PTB-XL.
5. **Report generation** — a template engine with clinical logic + LLM augmentation for the text (optional).
6. **Logging** — saving the case to a secure DB for audit and fine-tuning.

Data Model (Simplified)

Entity	Key fields
Patient	ID, age, sex
ECG record	ID, photo (hash), facility, date
Result	Record ID, patterns[], urgency, report text
User	ID, role (feldsher/physician/admin), facility

TS: Functional Requirements

FR-1: Image Upload and Preprocessing

- The user uploads a photo of the ECG strip via the camera or the gallery.
- The app automatically detects the strip borders and corrects perspective distortion.
- If the image quality is poor — a notification with instructions for retaking the photo.
- Supported formats: JPG, PNG, HEIC.

FR-2: ECG Digitisation and Analysis

- Signal extraction from 12 leads (or from the available leads if the image is partial).
- Automatic determination of heart rate, P-Q-R-S-T intervals, QRS duration, QTc.
- Pattern classification with the model's confidence indicated (confidence score).

FR-3: Results Display

- Result screen: urgency category (colour-coded), list of patterns, text report.
- Visualisation: digitised signal with abnormal segments highlighted.
- Action algorithm: 3–5 concrete steps (in Kyrgyz and Russian).
- Disclaimer: "The result is an auxiliary tool. The final decision rests with the physician."

FR-4: History and Logging

- Local storage of the last 50 cases on the device.
- Synchronisation with the server when internet is available.
- Export of the record to PDF for sending for consultation.

FR-5: User Management

- User registration linked to a facility (FAP/CFM).
- Roles: feldsher, GP, MoH administrator.
- Two-factor authentication for administrators.

FR-6: Administrative Panel (for the MoH KR)

- Dashboard: number of requests by region, pathology types, monthly trends.
- Management of facilities and users.
- Export of de-identified data for epidemiological analysis.
- Flags for manual verification by a cardiologist (model fine-tuning).

TS: Non-Functional Requirements, Security and Regulation

Performance

- Image processing time on the server: **< 10 seconds** (over a standard 3G internet connection).
- Offline lite model run time: **< 20 seconds** on an average Android device.
- Server availability: **99.5%** uptime.

Security and Confidentiality

- All patient data is transmitted over **TLS 1.3**.
- Medical data is stored in de-identified form (pseudonymisation in line with GDPR principles and the legislation of the KR).
- ECG photos are not stored on the server for longer than 30 days without the facility's explicit consent.
- Role-based access control (RBAC).

Regulatory Requirements for Implementation via the MoH KR

Requirement	Action
Medical software	Registration as a medical device with the MoH KR
Clinical validation	Pilot at the NCCIM: comparison of AI interpretations with expert ones — ≥100 cases
Informed consent	Patient consent protocol when data is stored
Algorithm audit	Quarterly report on model accuracy for the MoH
Interface languages	Kyrgyz + Russian (mandatory)

Model Limitations (Must Be Shown in the Interface)

- The model is trained on MIT-BIH + PTB-XL: data predominantly from a Western population. Fine-tuning on Kyrgyz data is required during the pilot phase.
- The quality of the result depends on the quality of the photo: lighting, shooting angle, resolution ≥ 5 MP.
- The system does not replace the physician — it is a **clinical decision support system (CDSS)**.

Implementation Roadmap

Phase 1 — Research and Prototype (Months 1–3)

- ☐ Study the MIT-BIH Arrhythmia Database (PhysioNet): data structure, annotations.
- ☐ Implement a basic pipeline: record loading → preprocessing → classification (CNN / BiLSTM).
- ☐ Implement a module for digitising paper ECGs from a photo (based on the open-source ECGtizer or an equivalent).
- ☐ Achieve ≥ 95% accuracy on the MIT-BIH test set across the 5 main classes.
- ☐ Document the methodology for inclusion in a school research paper.

Phase 2 — App MVP (Months 4–6)

- ☐ Develop the mobile interface (Flutter): photo upload, result screen, history.
- ☐ Deploy the backend API (FastAPI) on a cloud server.
- ☐ Integrate TensorFlow Lite for the offline mode.
- ☐ Translate the interface into Kyrgyz and Russian.

- ☐ Carry out internal testing on 20–30 ECG strips (collected from medical acquaintances).

Phase 3 — Pilot and Validation (Months 7–9)

- ☐ Approach the NCCIM or KSMA (Kyrgyz State Medical Academy) with a proposal for cooperation.
- ☐ Conduct pilot testing: 100+ cases, comparison with a cardiologist's expert interpretation.
- ☐ Collect feedback from feldshers at 2–3 FAPs (usability).
- ☐ Publish the research results (school conference / journal).

Phase 4 — Presentation to the MoH KR and Scaling (Months 10–12)

- ☐ Prepare a document package for the MoH: TS, pilot results, implementation plan.
- ☐ Present the project at a meeting of the MoH KR working group on digital health.
- ☐ Hand over the source code and documentation to the government partner.
- ☐ Apply for a grant (UNDP, WHO KR, AKF Health Program).

Expected Results and KPIs

Technical KPIs

Metric	Target value
Classification accuracy	≥ 97% on the MIT-BIH test set
Sensitivity (recall) for ACS	≥ 95% (critical: false negatives are unacceptable)
Specificity	≥ 93%
Photo processing time (server)	< 10 s
Offline model run time	< 20 s

Clinical and Social KPIs (Pilot)

Metric	Target value
Agreement with an expert cardiologist	≥ 90% on urgency category
Feldsher satisfaction (survey)	≥ 4/5 points
Number of facilities in the pilot	≥ 3 FAPs/CFMs
Number of ECGs processed during the pilot	≥ 100 cases

Portfolio Outcomes (for University Admission)

- A working prototype of the ECG-AI KG app with open source code on GitHub.
- A scientific publication or conference presentation (school / regional level).
- An official letter/MoU from the NCCIM or the MoH KR on cooperation.

- Pilot data: confusion matrix, ROC curve, feedback from medical staff.
- A project presentation for the MoH KR (this document is the basis).

Value for the MoH KR

The project addresses a problem that the MoH KR officially recognises as a priority: improving the quality of CVD diagnosis at the primary care level in remote districts. It does not require the purchase of new equipment — it uses the ECG machines and medical staff's smartphones that already exist. The cost of implementation in the first phase is minimal (cloud server + maintenance), while it can potentially cover all 980+ FAPs in the country.